

Week 1 Final Project Report

Adult Income Classification — Final Validation & Deployment Readiness

1. Problem Definition

The project predicts whether an individual earns more than \$50K/year using the UCI Adult/Census Income dataset. This is a binary classification task: 1 represents income above \$50K and 0 represents income at or below \$50K. The goal was to develop a reproducible end-to-end supervised-learning solution that can be evaluated on unseen data and reused for inference.

2. Data Preparation

A stratified 70% training, 10% development and 20% untouched test split was used with `random_state=42`. Missing values represented by '?' were handled in the preprocessing workflow. Feature engineering added age buckets, hours-per-week bins, a capital-gain indicator and log transform, a higher-education indicator, an education-hours interaction, and a married-status indicator. Numeric variables used median imputation and standardization; categorical variables used most-frequent imputation and one-hot encoding with `handle_unknown='ignore'`.

3. Model Development

Logistic Regression, Random Forest and HistGradientBoosting were evaluated during Week 1. Day 3 cross-validation identified HistGradientBoosting as the strongest shortlisted model on the main predictive metrics. Day 4 tuned Logistic Regression and HistGradientBoosting using reproducible RandomizedSearchCV with five-fold StratifiedKFold and precision as the primary metric.

4. Hyperparameter Tuning

Logistic Regression selected L1 regularization with $C=0.029763514416313176$ and achieved best CV precision of 0.7536. Its search space contained 40 combinations, so the requested 50 randomized iterations were automatically reduced to 40. HistGradientBoosting selected `max_leaf_nodes=31`, `max_iter=100`, `max_depth=3`, `learning_rate=0.02` and `l2_regularization=5.0`, with best CV precision of 0.8026. The higher tuned CV precision supported selecting HistGradientBoosting for the final workflow.

5. Calibration & Threshold Selection

The selected HistGradientBoosting pipeline was calibrated with sigmoid probability calibration using CalibratedClassifierCV. The classification threshold was selected on the development set by maximizing precision while maintaining recall of at least 0.50. The resulting operating threshold was 0.62.

6. Final Test Results

The calibrated final model was evaluated on the untouched test set. The observed results were:

Metric	Final Calibrated HistGradientBoosting
Accuracy	0.8535 (85.35%)
Precision	0.8042 (80.42%)
Recall	0.5128 (51.28%)
F1-score	0.6263 (62.63%)
ROC-AUC	0.9082 (90.82%)
PR-AUC	0.7778 (77.78%)
Brier score	0.1007
Threshold	0.62

The test classification report showed: $\leq 50K$ precision 0.86, recall 0.96 and F1 0.91 (support 7431); $> 50K$ precision 0.80, recall 0.51 and F1 0.63 (support 2338). Overall accuracy was 0.85 across 9769 test examples.

7. Error Analysis

The confusion matrix in the final notebook separates true negatives, false positives, false negatives and true positives. False positives are $\leq 50K$ individuals predicted as $> 50K$; false negatives are $> 50K$ individuals predicted as $\leq 50K$. The selected threshold favors precision, which is reflected in the 80.42% precision and lower 51.28% recall for the $> 50K$ class. Subgroup summaries for sex, race, education and marital status are included to identify possible performance differences.

8. Feature & Model Interpretation

The final notebook uses permutation importance on raw input variables and produces a feature-importance visualization. Development-stage analysis identified education-hours interaction, log-transformed capital gain, married-status information and age-related information as important predictive signals. These should be interpreted as predictive associations, not causal effects. The executed notebook contains the authoritative exact ranking.

9. Diagnostics

Learning-curve, ROC, precision-recall and calibration visualizations are generated by the notebook. These diagnostics are used to inspect generalization, discrimination and probability behavior rather than relying on accuracy alone.

10. Production Readiness

The final calibrated model and selected threshold are saved as `final_model.joblib`. The saved workflow contains preprocessing and feature transformations, so inference does not manually repeat imputation, scaling, encoding or feature engineering. The notebook reloads the artifact and tests it on 10 unseen examples, producing probabilities and thresholded predictions.

11. Documentation & Reproduction

The project includes `README.md`, `requirements.txt`, the executed final notebook, `final_model.joblib` and this report. To reproduce the project, install the listed dependencies, place `adult_dataset.csv` beside the notebook, and run the notebook from top to bottom. The notebook records the environment versions and generates the final evaluation and interpretation outputs.

12. Limitations & Future Improvements

The Adult dataset is historical and may not represent current populations. External validation on newer data is recommended. Before real-world use, subgroup/fairness analysis should be expanded, business-specific error costs should guide threshold selection, and monitoring for distribution drift and periodic retraining should be established. Additional models and features could be explored with more data and computational resources.

13. Conclusion

The Week 1 project now provides a complete workflow from data preparation through model tuning, calibration, threshold selection, final validation, interpretation, saved-artifact inference and reproducible documentation. The final calibrated HistGradientBoosting model achieved 85.35% accuracy, 80.42% precision, 51.28% recall, 62.63% F1, 90.82% ROC-AUC, 77.78% PR-AUC and a Brier score of 0.1007 on the untouched test set at a threshold of 0.62.